

02_Scope_of_Work

ALDRIDGE HARBOR PORT AUTHORITY

KESSLER POINT MARINE TERMINAL — PERIMETER SECURITY, ACCESS CONTROL AND SURVEILLANCE SYSTEMS

DOCUMENT 2 OF 13 — SCOPE OF WORK

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SENSITIVE SECURITY INFORMATION — 49 CFR PART 1520

ABBREVIATIONS AND ACRONYMS

Term	Meaning
ACS	Access Control System
AHPA	Aldridge Harbor Port Authority
AI	Artificial Intelligence
ARO	After Receipt of Order
ATP	Acceptance Test Procedure
BABA	Build America, Buy America Act
BOQ	Bill of Quantities
C&C	Command and Control
CCTV	Closed Circuit Television
CDR	Critical Design Review
CFR	Code of Federal Regulations
CISSP	Certified Information Systems Security Professional
COTP	Captain of the Port
CySO	Cybersecurity Officer
FAR	Federal Acquisition Regulation
FAT	Factory Acceptance Test
FEMA	Federal Emergency Management Agency
FSO	Facility Security Officer
FSP	Facility Security Plan
GCC	General Conditions of Contract
GICSP	Global Industrial Cyber Security Professional
GOS	Gate Operating System
IFB	Invitation for Bids
ISO	International Organization for Standardization
ITB	Instructions to Bidders
ITP	Inspection and Test Plan

MARSEC	Maritime Security (Level)
MTSA	Maritime Transportation Security Act
NFPA	National Fire Protection Association
NTP	Notice to Proceed
O&M	Operation and Maintenance
OCR	Optical Character Recognition
OSHA	Occupational Safety and Health Administration
OT	Operational Technology
PACS	Physical Access Control System
PDR	Preliminary Design Review
PIDS	Perimeter Intrusion Detection System
PTZ	Pan, Tilt, Zoom
RFI	Request for Information
SAT	Site Acceptance Test
SBOM	Software Bill of Materials
SOC	Security Operations Centre
SOW	Scope of Work
SSI	Sensitive Security Information
TOS	Terminal Operating System
TWIC	Transportation Worker Identification Credential
UPS	Uninterruptible Power Supply
USCG	United States Coast Guard
VMS	Video Management System

1. GENERAL

SOW-1.1 This Scope of Work describes the work to be performed, the personnel to be provided, the programme to be achieved, and the reporting and project management obligations of the Contractor.

SOW-1.2 The Contractor shall provide all design, engineering, labour, materials, equipment, tools, transport, supervision, testing, documentation and incidentals necessary for the complete and functional delivery of the Works, whether or not each item is expressly enumerated.

SOW-1.3 The Works shall be delivered at a facility in continuous commercial operation. The Contractor shall so plan and sequence the Works that terminal operations, vessel operations and gate operations are not interrupted.

SOW-1.4 The Facility Security Plan approved under 33 CFR 105.410 shall remain in continuous effect throughout the Contract Period. The Contractor shall not commence any activity which degrades a security measure described in that Plan until an amendment has been approved and compensating measures are in place.

SOW-1.5 The Contractor shall be responsible for all compensating security measures necessitated by its own works, including posted guards, temporary barriers and temporary lighting.

2. SCOPE OF THE WORKS

SOW-2.1 The Works comprise the following systems, each as detailed in Document 4:

Ref	System
A	Perimeter Intrusion Detection System — taut-wire detection fence to the full landside boundary, with patrol road
B	Video Surveillance System with artificial-intelligence object detection and classification
C	Security Pre-Gate facility, including Command and Control station
D	Electronic Access Control System, including pedestrian full-height turnstiles
E	Vehicle barriers and crash-rated gates
F	Security Operations Centre
G	Security network, structured cabling and outdoor field cabinets
H	Uninterruptible power supply and standby generation
I	Civil, electrical and builder's work in connection
J	Integration to the Terminal Operating System and to the Gate Operating System

SOW-2.2 The Commercial Gate and the Gate Operating System are provided by others and are outside this Contract. The Contractor's obligation is limited to the interfaces defined in Document 5.

SOW-2.3 The inter-gate queue lane civil works are provided by others.

SOW-2.4 The Contractor shall provide a complete and operational system. Where an item is necessary to render the Works complete and operational and is not expressly scheduled, it shall be deemed included.

3. KEY PERSONNEL

SOW-3.1 The Contractor shall assign the Key Personnel listed below. Curricula vitae shall be submitted with the bid in Package C.

Role	Minimum requirement	Commitment
Project Manager	Ten (10) years in electronic security systems projects, of which not fewer than five (5) years on marine terminal security projects	Full-time, on Site not fewer than three days per week
Technical Manager	Ten (10) years in electronic security systems integration	Full-time
Lead Designer	Seven (7) years in security systems design; Professional Engineer registration in the State, or equivalent	Not less than 50%
Cybersecurity Lead	Five (5) years in operational technology or industrial control system security; current CISSP, GICSP or equivalent certification	Not less than 25%
Commissioning Engineer	Five (5) years commissioning comparable systems	Full-time during Phases 5 and 6
Site Safety Officer	OSHA 30-hour construction certification; documented experience at a marine terminal	Full-time

SOW-3.2 Every member of the Key Personnel shall hold a valid Transportation Worker Identification Credential before mobilisation.

SOW-3.3 The Contractor shall not substitute any member of the Key Personnel without the prior written approval of the Authority.

SOW-3.4 A request for substitution shall be accompanied by the curriculum vitae of the proposed replacement, who shall possess qualifications and experience not less than those of the person being replaced.

SOW-3.5 Where the Contractor substitutes a member of the Key Personnel without prior written approval, or where the person named in the bid does not in fact perform the role from mobilisation, the Authority shall be entitled to liquidated damages of **twenty-five thousand United States dollars (\$25,000)** per occurrence. The Parties agree this sum is a genuine pre-estimate of the Authority's loss and not a penalty.

SOW-3.6 The Authority may at any time require the removal from the Works of any person whose performance or conduct it considers unsatisfactory. The Contractor shall comply within five (5) working days at its own cost.

4. SITE ACCESS AND ESCORT

SOW-4.1 All Contractor personnel requiring unescorted access to the secure area shall hold a valid Transportation Worker Identification Credential.

SOW-4.2 Personnel not holding a valid credential shall be escorted at all times by a credential holder, at the ratios set out in the Facility Security Plan.

SOW-4.3 The Contractor shall be responsible for providing escorts.

SOW-4.4 The Contractor shall submit the names of all personnel requiring Site access not fewer than ten (10) working days before first attendance.

SOW-4.5 Site working hours are 07:00 to 17:00 Monday to Friday. Work outside these hours, and any work on a Saturday, Sunday or public holiday, requires the prior written approval of the Authority, which shall not be unreasonably withheld.

5. PROGRAMME AND MILESTONES

SOW-5.1 The Works shall achieve Final Acceptance within **eighteen (18) months** of the date of Receipt of Order (ARO).

SOW-5.2 The Contractor shall achieve the following milestones:

Milestone	Requirement	Due from ARO
M1 — Kick-off Meeting	At the Site, with all Key Personnel present	14 days
M2 — Baseline Programme submitted	Resource-loaded, critical path identified	21 days
M3 — Preliminary Design Review (PDR)	Design intent, system architecture, equipment selections, preliminary drawings. Authority approval is a precondition to the placing of any procurement order.	60 days
M4 — Critical Design Review (CDR)	Detailed design, final drawings, calculations, schedules, cable schedules, method statements. Authority approval is a precondition to commencement of installation.	150 days
M5 — First equipment delivery	Batch 1, per SOW-6	210 days
M6 — Head-end operational	Phase 1 complete, Security Operations Centre functional	300 days
M7 — Perimeter complete	All sectors installed and detecting	390 days
M8 — Pre-Gate cutover	Security pre-gate operational, old arrangement withdrawn	450 days
M9 — Site Acceptance Test complete	All systems, witnessed	510 days
M10 — Training complete	All courses delivered	525 days
M11 — Final Acceptance	All punch-list items closed, as-builts and O&M manuals accepted	540 days

SOW-5.3 The Contractor shall submit a resource-loaded baseline programme in Primavera P6 format, showing the critical path, not later than milestone M2.

SOW-5.4 The Contractor shall update the programme monthly and shall report all variances against the baseline together with proposed recovery measures.

SOW-5.5 Time is of the essence.

6. SUPPLY

SOW-6.1 Equipment shall be delivered in batches aligned to the sector release schedule issued by the Authority.

SOW-6.2 The Authority will issue sector releases as terminal operations permit.

SOW-6.3 The Contractor shall provide secure, weatherproof storage for all delivered equipment. The Authority will not provide storage.

SOW-6.4 The Contractor shall submit a procurement schedule at PDR identifying all long-lead items, their lead times and their required order dates.

SOW-6.5 The Contractor shall bear the risk of all equipment until Final Acceptance.

7. DESIGN

SOW-7.1 The Contractor shall prepare and submit for approval a complete design, comprising as a minimum: system architecture drawings; site layout drawings showing every device location; single-line diagrams; cable schedules and routing drawings; device schedules; power calculations; heat load calculations; storage calculations; bandwidth calculations; camera field-of-view calculations demonstrating compliance with the detection and classification requirements of Document 4; structural calculations for all poles, masts and foundations; and method statements for every installation activity.

SOW-7.2 Design submissions shall be made in native format and in PDF.

SOW-7.3 The Authority will review and respond to design submissions.

SOW-7.4 The Contractor shall not proceed on the basis of an unapproved design. Work performed on an unapproved design is at the Contractor's sole risk and may be ordered removed at the Contractor's cost.

SOW-7.5 Approval of a design by the Authority shall not relieve the Contractor of any obligation under the Contract nor of responsibility for the adequacy of its design.

8. INSTALLATION

SOW-8.1 Installation shall proceed sector by sector in accordance with the approved phasing plan.

SOW-8.2 Not more than one perimeter sector shall be open at any time.

SOW-8.3 Where a perimeter sector is open, the Contractor shall provide continuous manned guarding of that sector until the new detection line is operational.

SOW-8.4 The Contractor shall maintain the existing security systems in full operation until the replacement system for the affected area has been accepted.

SOW-8.5 All works shall comply with NFPA 70, NFPA 731, and the standards scheduled at Annex B.

SOW-8.6 The Contractor shall keep the Site clean and shall remove all waste daily. Waste shall not accumulate in the operational yard.

9. TESTING AND ACCEPTANCE

SOW-9.1 The Contractor shall submit an Acceptance Test Procedure for approval not later than CDR.

SOW-9.2 A Factory Acceptance Test shall be performed on the video management system, the access control system and the security management platform prior to shipment. The Authority may witness.

SOW-9.3 A Site Acceptance Test shall be performed on the complete integrated system. The Authority will witness the Site Acceptance Test.

SOW-9.4 The Site Acceptance Test shall demonstrate, as a minimum: detection performance of the perimeter system against the probability of detection and nuisance alarm rate requirements of Document 4, measured over a continuous **thirty (30) day** soak period; classification accuracy of the video analytics; end-to-end transaction time at the pre-gate; failover of all redundant elements; and full operation on standby power.

SOW-9.5 Where any test fails, the Contractor shall rectify and the test shall be repeated in full. The Authority's costs of witnessing a repeated test shall be borne by the Contractor.

SOW-9.6 Substantial Completion shall be certified when the Works are capable of being used for their intended purpose and all life-safety and security functions are operational.

SOW-9.7 Final Acceptance shall be certified when all punch-list items are closed, all documentation has been accepted, all training has been delivered, and the system has operated for **ninety (90) consecutive days** without a Severity 1 or Severity 2 fault.

10. TRAINING

SOW-10.1 The Contractor shall deliver training to the following audiences:

Course	Audience	Duration
Operator	Security Operations Centre operators and pre-gate operators	5 days
Administrator	System administrators, access control administrators	5 days
Maintainer	Authority maintenance technicians	10 days
Cybersecurity	Authority IT and the Cybersecurity Officer	3 days

SOW-10.2 Training will be delivered at the Site in English.

SOW-10.3 The Contractor will provide training materials for each course, and may provide these in electronic form.

SOW-10.4 Each course shall be delivered not fewer than twice, to accommodate shift patterns.

SOW-10.5 Training shall be of a standard satisfactory to the Authority.

SOW-10.6 The Contractor will provide a recorded version of each course for the Authority’s subsequent use.

11. DOCUMENTATION AND HANDOVER

SOW-11.1 The Contractor shall deliver as-built drawings in native format and PDF within thirty (30) days of Site Acceptance Test completion.

SOW-11.2 The Contractor shall deliver Operation and Maintenance manuals covering every item of supplied equipment, including manufacturer’s literature, maintenance schedules, spare parts lists with part numbers and recommended holdings, and troubleshooting procedures.

SOW-11.3 The Contractor shall deliver a complete set of software licences registered in the name of the Authority, together with all licence keys, administrative credentials and configuration backups.

SOW-11.4 The Contractor shall deliver a Software Bill of Materials for every software component supplied, in CycloneDX or SPDX format.

SOW-11.5 The Contractor shall deliver a complete configuration backup of every system at Final Acceptance and at each subsequent anniversary during the warranty period.

SOW-11.6 All documentation containing Sensitive Security Information shall be marked and handled in accordance with Annex C.

12. WARRANTY AND MAINTENANCE

SOW-12.1 The Contractor shall warrant the Works for **thirty-six (36) months** from the date of Final Acceptance.

SOW-12.2 During the warranty period the Contractor shall provide corrective maintenance for all defects, and preventive maintenance in accordance with the manufacturers’ schedules.

SOW-12.3 The Contractor shall respond to faults within the following times:

Severity	Definition	Response	Resolution
1	Total loss of a system, or loss of a security function required by the Facility Security Plan	2 hours	8 hours
2	Partial loss affecting more than 10% of a system	4 hours	24 hours
3	Single device failure	24 hours	5 working days
4	Cosmetic or non-functional	5 working days	30 days

SOW-12.4 The Contractor shall maintain all system software at a currently supported release, and shall apply security patches throughout the warranty period.

SOW-12.5 The Contractor shall hold a spares holding on Site sufficient to meet the resolution times at SOW-12.3.

SOW-12.6 The Contractor will provide remote support access, subject to the Authority’s cybersecurity requirements at Document 6.

SOW-12.7 At the expiry of the warranty period the Contractor shall hand over all spares remaining in the Site holding to the Authority at no additional cost.

13. REPORTING AND PROJECT MANAGEMENT

SOW-13.1 The Contractor shall submit a monthly progress report not later than the fifth (5th) working day of each month, containing: progress against the baseline programme; a look-ahead of not less than eight weeks; the status of all submittals and requests for information; procurement status of all long-lead items; quality and safety statistics; a risk register with owners and mitigations; and photographic records.

SOW-13.2 The Contractor shall attend a weekly progress meeting at the Site, chaired by the Authority.

SOW-13.3 The Contractor shall maintain a submittal register and a request-for-information register in a format agreed with the Authority.

SOW-13.4 The Contractor shall respond to any Authority query within **five (5) working days**.

SOW-13.5 The Contractor shall maintain a quality management system in accordance with ISO 9001 and shall submit inspection and test plans for all

major activities.

SOW-13.6 The Contractor shall submit a Health and Safety Plan specific to the Site before mobilisation, addressing the hazards of working within an operating container terminal, including container-handling equipment movements, rail movements, and work adjacent to the quay edge.

SOW-13.7 The Contractor shall report every accident, incident and near-miss to the Authority within twenty-four (24) hours.

SOW-13.8 The Contractor shall report any cybersecurity incident affecting the Works to the Authority immediately upon becoming aware of it.

END OF DOCUMENT 2